

LibChat Transcript Analysis



Using machine learning in Python to analyze chat data

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Field Experience

- As an MLIS student, completing a Field Experience is a way to obtain real world, professional experience prior to graduating.
- Achieving the ADDM Certificate (Data Science) has been my area of focus for the program.
- I spent this summer semester working alongside the assessment team (Berenika, Jenn, Anne, and Diana), seeking to experiment with using machine learning to analyze chat transcripts pulled from LibChat interactions over the past 6 years.

Project Introduction

Our Goal	Timeline	Methods
Create a machine learning model that will: • Help assess strong/weak points in our LibChat service • Guide future trainings/decision-making • Is repeatable for future applications.	• 130-Hours over the course of 12 weeks. • 1-hour weekly check-ins to denote progress and receive feedback	Model 1: Predictive Model using Linear Regression Model 2: Predictive Binary Classification Model using Sentiment Analysis Model 3: Feature Extraction Model 4: Split Chat and Topic Extraction

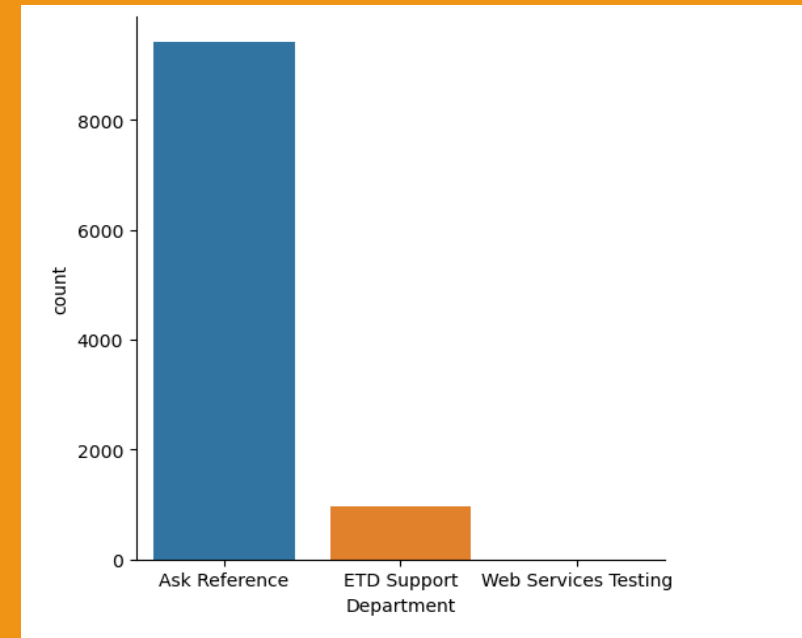
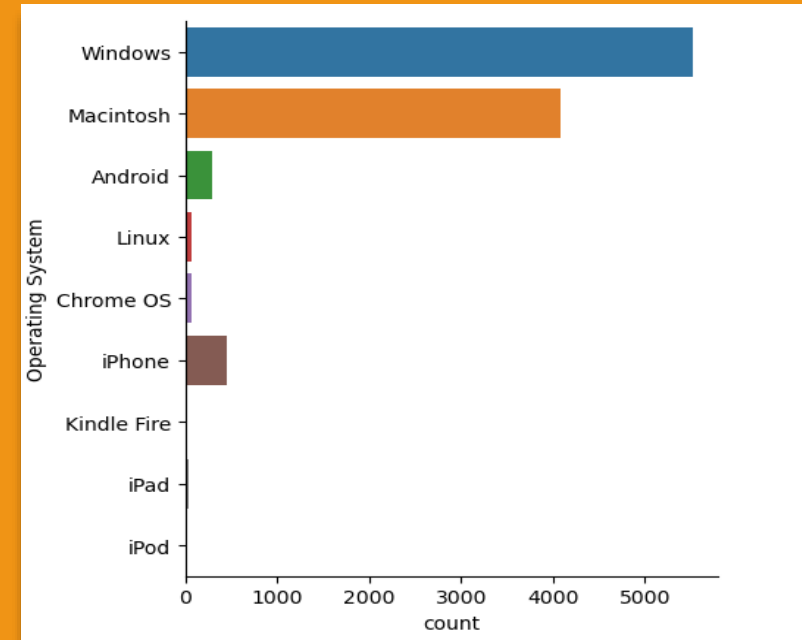
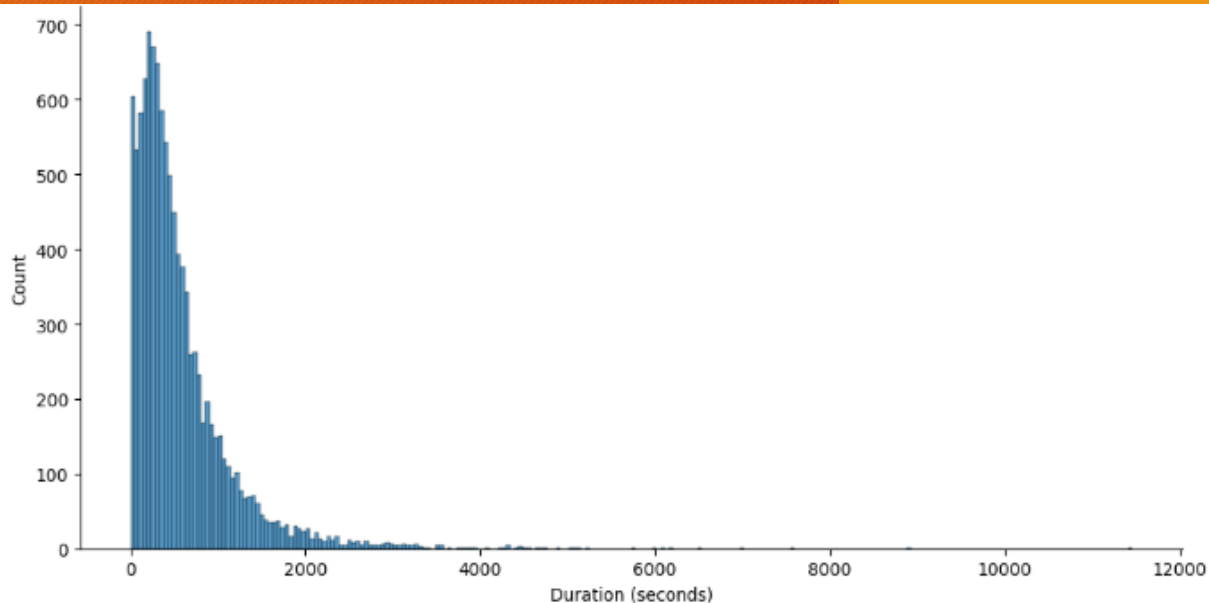
Variable Name	Description	Data Type	Variable Name	Description	Data Type	
Chat ID	Unique ID # of chat	integer	Transfer History	If interaction was transferred mid-chat	object	
Browser	Description of user’s search engine	object		Message Count	Number of messages in interaction	integer
Operating System	Machine of user	object	Internal Note		Note made only viewable by staff	object
User Agent	Info about user’s machine	object		Tags	N/A	object
Referrer	URL of user at time of interaction	object	Ticket ID		Unique ID # of ticket created by staff	float
Widget	Widget used to initiate interaction	object		Deidentified_Answerer	Anonymized LibChat staff member	object
Department	Type of interaction	object	Deidentified_Email		Anonymized user	object
Timestamp	Time/date	object		Deidentified_Transcript	Anonymized chat interaction	object
Wait Time (s)	Time waited in seconds	integer	Deidentified_Initial_Question		Anonymized first question asked	object
Duration (s)	Time elapsed during interaction in seconds	integer				
Screensharing	If screen was shared	object				
Rating (0-4)	User rating of chat	integer				
Comment	Comment left by staff	object				

Exploratory Data Analysis

- In any data project, the first step is to explore the data.
 - Count number of rows & columns (shape of the data)
 - Count number of unique values
 - Check the data types of each column in the data
 - Check for missing data
 - Visualize counts, distributions, and combinations

EDA Highlights

- Let's look at a few examples of what our Exploratory Data Analysis can show us.



Stats by LibChat Staff

EDA Highlights

When we originally looked at the data and the averages for staff, these are the numbers that were produced.

I want to use this example to emphasize the importance of exploring your data before starting any research project.

	Deidentified_Answerer	count	Wait Time (seconds)	Duration (seconds)	Rating (0-4)	Message Count
0	Librarian 41	777	15.411840	534.402831	1.109395	13.487773
1	Librarian 49	537	10.985102	644.378026	1.124767	17.141527
2	Librarian 60	440	8.834091	322.131818	1.211364	12.763636
3	Librarian 12	437	12.109840	445.592677	1.295195	13.361556
4	Librarian 4	421	7.748219	509.641330	1.118765	13.429929
5	Librarian 39	395	8.412658	452.582278	1.354430	12.321519
6	Librarian 82	341	32.266862	704.510264	1.322581	13.445748
7	Librarian 5	296	14.547297	1059.391892	1.368243	23.942568
8	Librarian 38	291	10.189003	437.587629	1.542955	12.343643
9	Librarian 27	278	8.064748	561.237410	0.712230	13.115108
10	Librarian 19	278	11.931655	847.269784	1.190647	17.906475
11	Librarian 8	268	14.798507	591.660448	1.190299	18.175373
12	Librarian 58	251	22.645418	1249.617530	1.633466	21.258964
13	Librarian 50	224	9.058036	420.812500	1.147321	15.866071
14	Librarian 23	224	10.151786	586.825893	0.750000	7.928571
15	Librarian 14	220	15.236364	588.213636	0.913636	14.804545
16	Librarian 70	212	10.000000	409.806604	1.264151	10.896226
17	Librarian 56	207	20.632850	450.241546	1.251208	10.328502
18	Librarian 9	199	11.361809	348.924623	0.979899	11.085427
19	Librarian 10	180	7.533333	476.877778	0.755556	13.211111

EDA Highlights

Stats by LibChat Staff

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	Deidentified_Answerer	count	index	Wait Time (seconds)	Duration (seconds)	Rating (0-4)	Message Count
0	Librarian 41	223	7059.874439	7.551570	575.233184	3.865471	16.000000
1	Librarian 49	154	7204.402597	10.032468	650.233766	3.922078	19.584416
2	Librarian 12	144	5041.402778	14.062500	521.263889	3.930556	15.972222
3	Librarian 39	136	6375.838235	8.448529	488.088235	3.933824	14.139706
4	Librarian 60	136	7110.382353	7.514706	376.022059	3.919118	15.625000
5	Librarian 4	121	2425.520661	7.404959	576.793388	3.892562	15.371901
6	Librarian 82	118	9163.237288	19.745763	879.516949	3.822034	15.805085
7	Librarian 38	118	6714.813559	9.457627	442.728814	3.805085	14.050847
8	Librarian 58	105	7397.171429	17.933333	1546.438095	3.904762	26.247619
9	Librarian 5	102	5679.225490	12.558824	1250.549020	3.970588	28.833333
10	Librarian 19	86	5639.348837	17.441860	965.848837	3.848837	20.360465
11	Librarian 8	82	4593.634146	12.231707	661.975610	3.890244	20.804878
12	Librarian 70	70	7365.757143	9.171429	445.357143	3.828571	12.671429
13	Librarian 50	68	5824.279412	8.544118	468.750000	3.779412	17.661765
14	Librarian 56	66	7385.878788	14.257576	431.424242	3.924242	12.439394
15	Librarian 27	52	4431.442308	5.673077	653.365385	3.807692	17.923077
16	Librarian 14	51	6402.313725	14.294118	602.901961	3.941176	15.529412
17	Librarian 75	50	8386.140000	7.080000	631.180000	3.860000	17.360000

EDA Highlights

Here we can see the same effect on the user averages.

In addition, we now see just how many interactions have no data for the user email. This information can help us later or in other projects.

Stats by User

	Deidentified_Email	count	Wait Time (seconds)	Duration (seconds)	Rating (0-4)	Message Count
0	No Email Provided	1137	20.218118	847.448549	1.189974	17.360598
1	User0b69e697c9@pitt.edu	38	22.263158	561.684211	3.473684	18.815789
2	User5b0d45d71d@pitt.edu	36	8.222222	525.555556	0.333333	15.750000
3	User4ade0d051b@pitt.edu	36	9.111111	222.111111	1.472222	6.027778
4	Userce43e555d3@pitt.edu	29	7.034483	544.103448	0.379310	16.172414
5	User03c21342b9@pitt.edu	21	8.380952	422.571429	0.904762	9.571429
6	User140bd3eabb@pitt.edu	21	17.047619	514.809524	0.571429	10.619048
7	User10845390e0@pitt.edu	20	8.200000	705.850000	0.000000	14.600000
8	User5a21cf4166@pitt.edu	19	9.736842	534.052632	2.421053	18.842105
9	User99c429d285@pitt.edu	19	7.210526	403.894737	1.368421	12.842105
10	Userf87c6c097e@pitt.edu	19	5.105263	644.210526	0.210526	20.421053
11	User48db1f6412@pitt.edu	19	7.947368	667.631579	1.684211	17.684211
12	User6b5901f44b@pitt.edu	18	9.833333	620.166667	0.444444	17.000000
13	User5b1a29a6ab@pitt.edu	17	6.882353	718.117647	1.647059	27.764706
14	User23119a5db6@pitt.edu	16	8.312500	475.000000	1.312500	12.125000
15	User56bfb53246@pitt.edu	16	6.625000	473.500000	0.500000	8.500000
16	User2103908f3b@pitt.edu	15	14.000000	559.333333	0.533333	10.866667
17	User6e95143f51@pitt.edu	15	8.666667	580.733333	2.133333	14.533333
18	User456f1b6207@pitt.edu	15	29.133333	433.600000	0.000000	13.000000
19	User51175ec644@pitt.edu	15	23.066667	751.000000	1.533333	12.733333

EDA Highlights

Here we can see the same effect on the user averages.

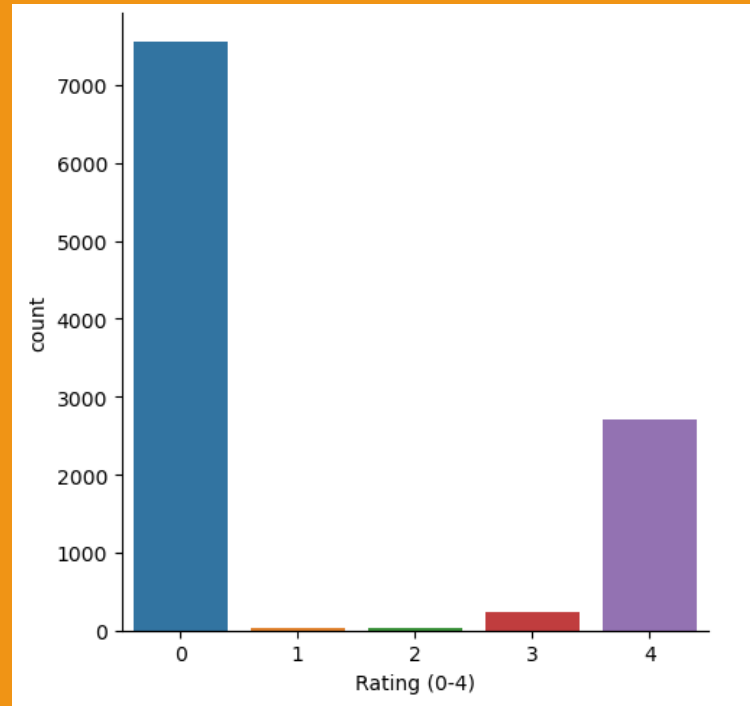
In addition, we now see just how many interactions have no data for the user email. This information can help us later or in other projects.

Stats by User

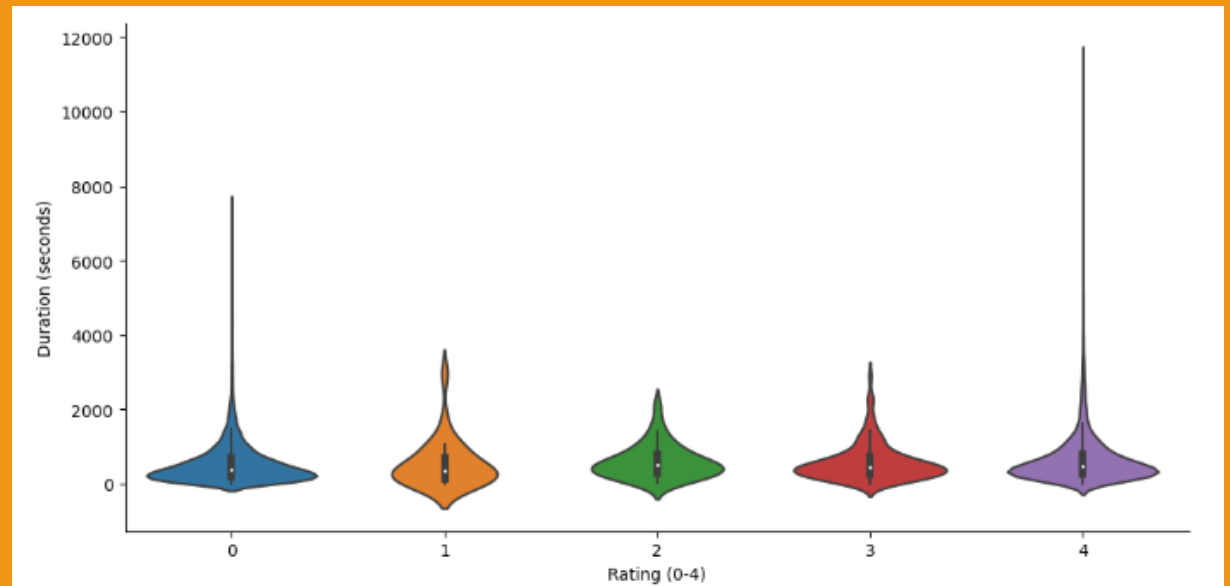
	Deidentified_Email	count	index	Wait Time (seconds)	Duration (seconds)	Rating (0-4)	Message Count
0	No Email Provided	351	7376.105413	17.435897	1157.766382	3.854701	23.014245
1	User0b69e697c9@pitt.edu	33	6478.545455	21.121212	592.575758	4.000000	19.848485
2	User4ade0d051b@pitt.edu	14	7467.000000	11.142857	417.571429	3.785714	9.000000
3	Userf9e1d818b1@pitt.edu	13	6706.153846	14.153846	420.923077	3.769231	14.307692
4	User5a21cf4166@pitt.edu	12	6201.333333	10.666667	512.500000	3.833333	20.500000
5	User136833aaa2@pitt.edu	9	5425.888889	11.111111	604.333333	3.888889	17.222222
6	Userf40fd6dbb6@pitt.edu	9	4200.111111	8.444444	324.444444	4.000000	12.777778
7	Usa69048cda0@pitt.edu	9	6310.666667	16.111111	435.666667	4.000000	15.000000
8	User6e95143f51@pitt.edu	8	4871.250000	9.500000	503.375000	4.000000	14.125000
9	User0a2342546b@pitt.edu	8	3394.125000	12.125000	565.500000	4.000000	15.625000
10	User48db1f6412@pitt.edu	8	7245.125000	6.625000	899.750000	4.000000	24.625000
11	User99c429d285@pitt.edu	7	7135.285714	6.857143	522.714286	3.714286	12.571429
12	User36fa7664be@pitt.edu	7	5366.571429	4.000000	814.857143	4.000000	23.714286
13	User5b1a29a6ab@pitt.edu	7	1650.285714	6.714286	513.142857	4.000000	18.285714
14	User2a5e302cec@pitt.edu	7	4208.142857	6.428571	660.142857	3.714286	15.857143
15	User51175ec644@pitt.edu	6	7166.833333	43.000000	1080.500000	3.833333	20.166667
16	Userc8c7698e51@pitt.edu	6	7289.500000	25.000000	443.833333	3.833333	11.333333
17	User05ad487ad5@pitt.edu	6	7441.166667	9.166667	486.500000	4.000000	18.666667

EDA Highlights

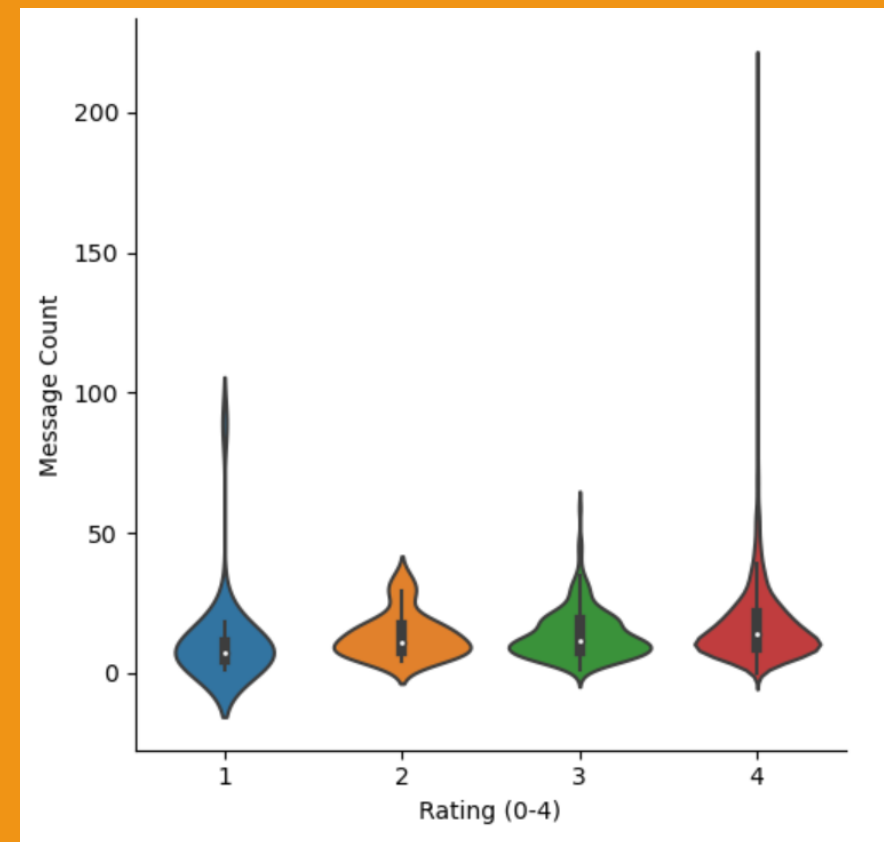
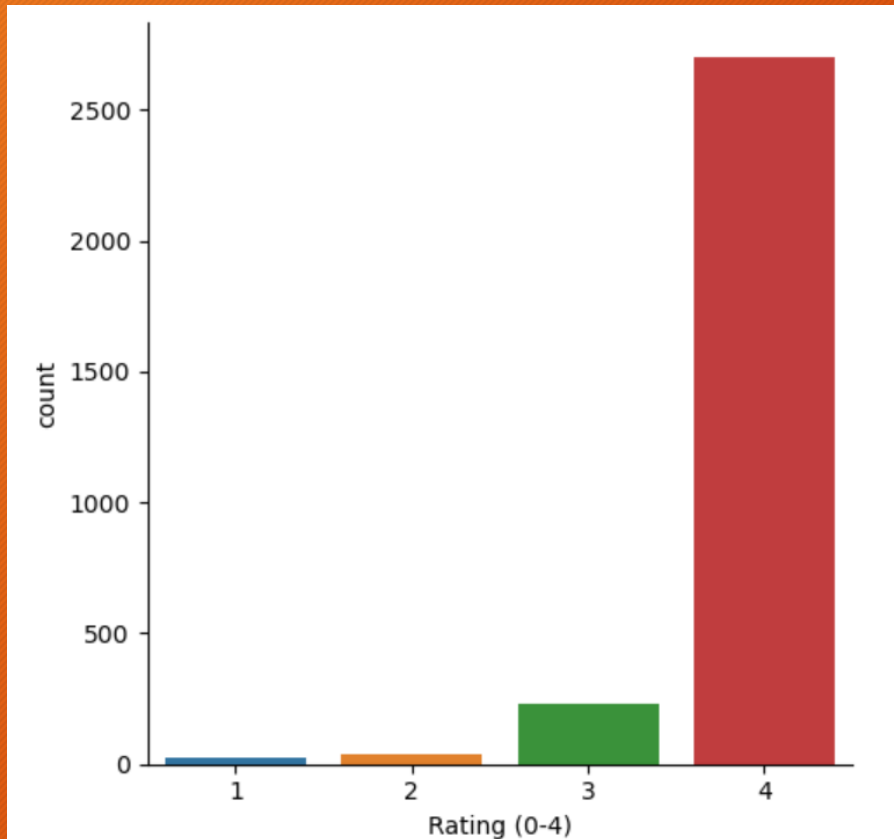
- The Rating column is what we want to use as our output variable, so let's explore that a bit more in depth.
- Using the Duration variable as an example, we can check to see if the shape of the "0" or unrated chats appear like any of the shapes of the rated chats.



[Here we can see how many unrated chats overwhelm our dataset]



EDA Highlights



- As stated in our original goals, we want to focus on how we can predict ratings of a given chat. For that, we must tidy our data and focus on chats only with ratings.
- We will keep in mind that we are trimming A LOT of data points, but we must work with what we have for now.

Model 1: Predictive Model using Linear Regression



In our first approach, we attempted to predict patrons' ratings using machine learning, with an output variable mimicking our previous 1-4 rating system.



Linear Regression is a statistical technique that uses a linear equation to predict the value of a variable (in our case Ratings) based on the value of another variable(s).



Using the data we extracted, I wrote several “models” implementing Python machine learning libraries such as statsmodels, scikit-learn, and RandomForest.

Model 1: Predictive Model using Linear Regression Results

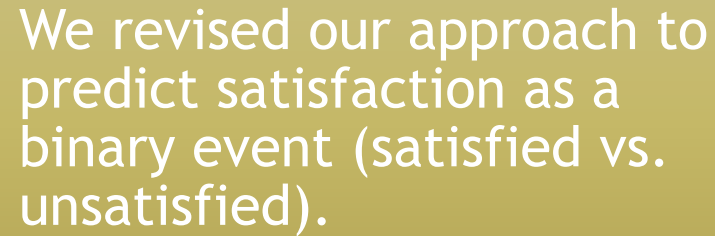
- This model proved ineffective due to the subjective nature of ratings, which only comprised about 18% of our data.
- Using statistical scoring metrics, we were able to identify that this approach was no more effective at predicting chat ratings than using the average of each variable.

my reaction to that information



Model 2: Predictive Binary Classification Model using Sentiment Analysis

We revised our approach to predict satisfaction as a binary event (satisfied vs. unsatisfied).



```
graph TD; A[We revised our approach to predict satisfaction as a binary event (satisfied vs. unsatisfied).] --> B[Ratings of 1-3 were categorized as "Unsatisfied," and a rating of 4 as "Satisfied."]; B --> C[I then used similar modeling techniques to model 1, implementing the sentiment scores as additional input variables.];
```

Ratings of 1-3 were categorized as "Unsatisfied," and a rating of 4 as "Satisfied."

I then used similar modeling techniques to model 1, implementing the sentiment scores as additional input variables.

Model 2: Predictive Binary Classification Model using Sentiment Analysis Results

- While this model yielded clearer results, it revealed that positive sentiment alone did not always correlate with higher ratings.
- Neutral language often correlated with lower ratings, but the model still did not provide the desired predictive power.



Model 2: Predictive Binary Classification Model using Sentiment Analysis Results

- Example of a 1-Rated chat with a highly positive sentiment score:

```
I have a book out that is due today; however, I'm out of town for Spring Break. Can you grant me a week extension? Thank you.
-[nameredacted]
10:09:13 - Userf9e1d818b1 : Hi, [nameredacted].
10:09:54 - Librarian 38 : Hello, I cannot renew your items. I'll have to create a help ticket and someone will get to you directly. Can you provide the best email address to reach you?
10:10:48 - Userf9e1d818b1 : [emailredacted]
10:10:57 - Librarian 38 : Excellent, someone will get back to you soon!
10:10:57 - Userf9e1d818b1 : I'll keep an eye out for an email then.
10:11:04 - Userf9e1d818b1 : Thanks! Enjoy Break!
10:11:04 - Librarian 38 : Please ULSMember not hesitate to ask us if you have any additional questions. Have a great day!
```

neg	0.024969
neu	0.175865
pos	0.799166

Model 2: Predictive Binary Classification Model using Sentiment Analysis Results

- Example of a 4-Rated chat with a highly negative sentiment score:

Can you please pass this onto somebody who can get some staplers in Hillman? I'm pretty disappointed that I'm paying \$99,999 in tuition this year to have barely any access to university resources, and I can't even get my hands on a stapler when I go to Hillman for the 9 hour time frame that it's open. This is ridiculous. 14:22:59 - Librarian 49 : [nameredacted] [nameredacted], thanks for writing in.

neg	0.936330
neu	0.056892
pos	0.006779

Model 3: Feature Extraction and Topic Modeling



We utilized natural language processing to extract key topics from chat transcripts. This model successfully identified language associated with higher and lower ratings.



Positive interactions often included phrases like “Thank you very much,” while negative ones included “don’t,” “cannot,” or “don’t know.”



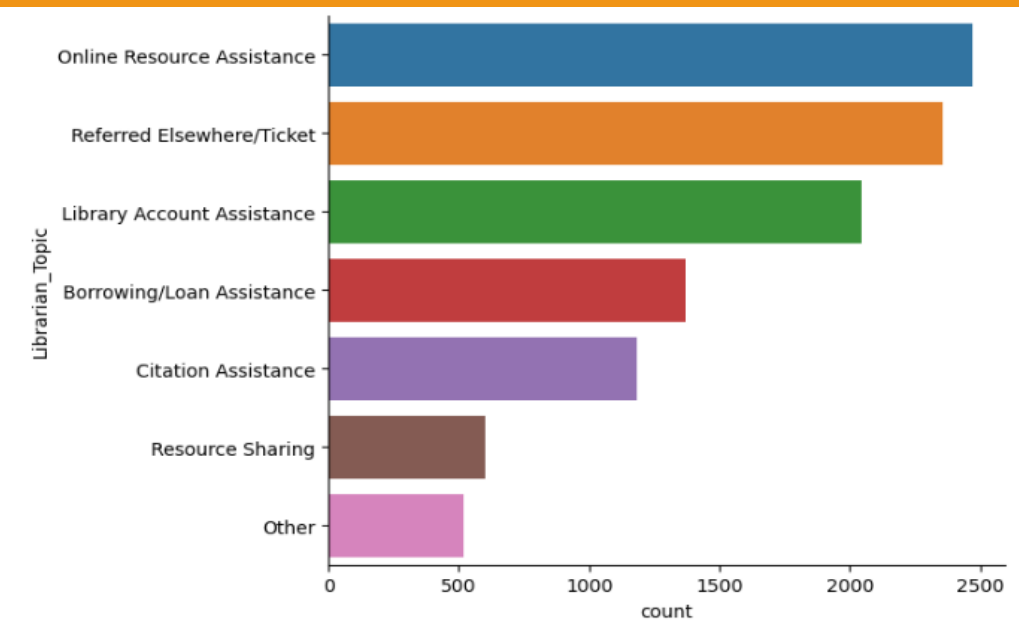
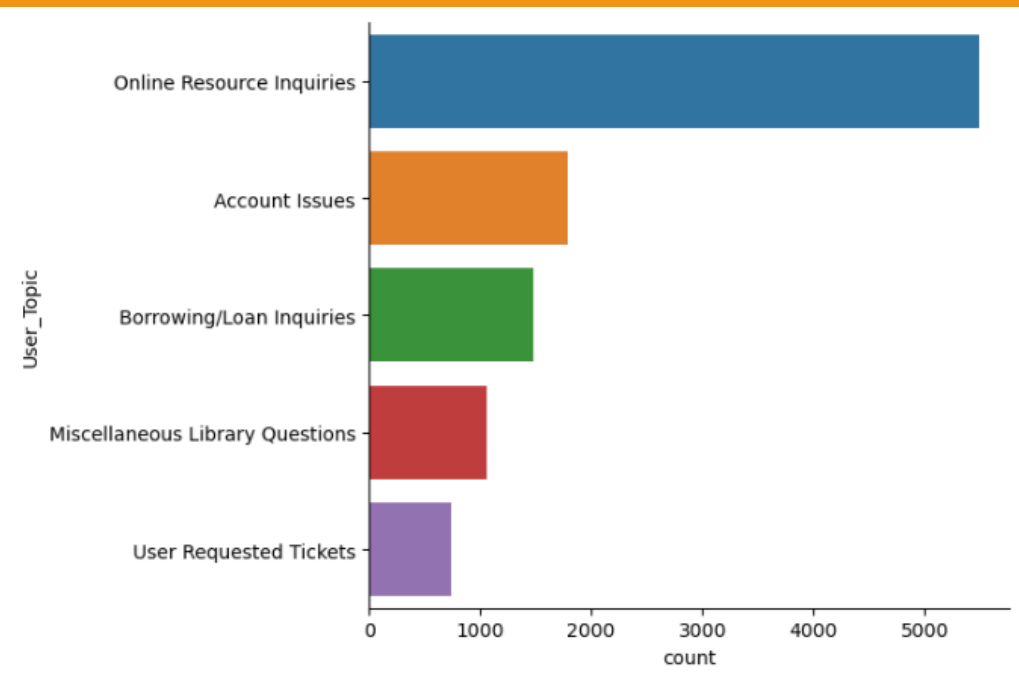
However, this model was initially designed for review analysis, not raw transcript analysis. Applying it to review comments, as we plan to do with our NPS scoring system, could yield more significant results.

Model 4: Split Chat and Topic Extraction

This model, our most refined, used natural language processing, TFIDF vectorization, and regular expressions to split chats into “Librarian” and “User” segments. We aimed to:

1. Analyze “question” and “answer” topics.
2. Identify differences in language between users and librarians.
3. Assess strengths and weaknesses based on these topics.

Key Findings: The Topics



Key Findings: Questions & Answers

The main takeaway of this goal was how we handled certain questions. This point led to my final conclusions and helped create a clear theme for our topics. What our patrons sought out help for on LibChat most was for assistance with online resources. This includes but is not limited to; inquiring about access to online databases/journals, seeing if the library had digital access to a print text, troubleshooting errors with online resources, etc.

The most common response to these inquiries was two-fold. Staff would do their best to assist in real time, providing helpful information that fell under the “Online Resource Assistance” topic. The other half of the time, these inquiries were “Referred Elsewhere”.

Key Findings: Library term differences

When speaking about general borrowing terms, the language of users and librarians varied. The biggest example was the term “reserve”. This term was used widely by users to refer to a variety of inquiries but mainly to seek how to place a hold on a particular item. The divide with this term is that librarian side of chat mainly used this term to refer to “course reserves”.

Key Findings: Our Strengths

Tying into the previous goal, we began to see a pattern when we look at the intersection of user questions and librarian answers. The overwhelming majority of patron inquiries were handled across several librarian topics, as we saw a lot of overlap in patron questions that required a diversity of answers.

According to the data, where we were the strongest was on examples of loan information and general borrowing inquiries. This includes but is not limited to; renewing items, placing holds on patron's behalf's, instructing patrons on how to request from other institutions (Resource Sharing) and explaining loan policies.

Key Findings: Our Weakness

Where the biggest room for improvement our research showed was in finding experts for more nuanced questions. In essence, the data will show that a major part of the inquiries that were sent to LibChat needed to be redirected in some capacity to the relevant expert.

Out of close to 11,000 interactions, less than 2,000 tickets were created, less than 19%.

With this being the case, the data points us to a significant gap in coverage for inquiries that cannot be easily answered in chat. An educated assumption based on some of the chat analysis is that in the past often a person's email, phone number, or location was given to the patron instead of a ticket being created on their behalf. The patron then would have to reach out to the relevant expert on their own with little to know explanation of a timeline. This also makes the patron reiterate all their concerns they just spent the time doing on that given interaction.

Key Findings: The Context

Our “Unsatisfied” ratings (Ratings below a 4) are miniscule in number compared to our 4 rated chats even when they are all combined. Our rated chats in general only make up 23% of the data.

I believe leaning into our strong suits and emphasizing trainings on our weak points can lead to more satisfied interactions and in turn more ratings overall.

Key Findings: Justification for Change

As previously mentioned, the Assessment unit has moved on from this way of evaluating chats (on a simple 1-4 survey). Instead, they have begun utilizing, since the beginning of 2024, NPS Scoring which gives users a 1-10 scale and the option to leave verbose comments/reviews.

With this switch, we can implement some of the modeling used in this project to further predict and analyze performance, giving us the ability to compare the two rating systems to each other.

Wrap-up

- In conclusion, our LibChat team has been doing phenomenal work for several years and the data speaks to exactly that.
- In such a subjective topic as ratings, it is hard to pinpoint exactly what makes an individual satisfied vs unsatisfied with an interaction.
- However, I hope through this project we can see the usefulness in machine learning as a tool for advancement and the wide variety of applications and use-cases it can have here at the ULS and in the library field.